AI for SARS

Future Readiness, Research, and Responsible AI

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DAY III

Content

Your Request for Some Action!

Content

Content

Content

Content

Content

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Scan to Access LandBot

Day 2 Recap

Recap — Painpoints & AI Mapping

- **Painpoints Identified**
  - Operational bottlenecks + risk gaps
  - Compliance inefficiencies + governance concerns

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- **AI-Driven Solutions Mapped**
  - Clear linkage: painpoint → AI-driven solutions
  - Defined impact: efficiency, risk reduction, trust

Recap — Co-Creation & Capability Momentum

- **Co-Created Solutions**
  - Designed by SARS team (not imposed externally)
  - Shared views of future direction

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- **Emerging Technical Appetite**
  - Interest in deeper model, tooling, and architecture engagement
  - Momentum toward sustained AI capability development

Team Sakkie — Pain Point & Proposed Solution

Pain Point

- No single-view taxpayer dashboard.
- Fragmented visibility across channels.
- Slow first-time resolution and follow-ups.

Proposed AI Solution

Team Members: Maemu (Lead), Derek, Molatelo, Esther

- Single-view AI dashboard for taxpayer status + history.
- Countdown timers and next-best-action prompts for cases.
- **Success:** fewer handoffs, faster resolution, improved taxpayer trust.

Team The Pioneers — Pain Point & Proposed Solution

Pain Point

- Audit findings overturned due to missed steps.
- Weak process governance reduces defensibility.
- Rework increases audit cycle time.

Proposed AI Solution

Team Members: Ramakrishan Pasungili (Lead), Ivonne Vorster, Karen Terblanché, Khutso Nong

- AI-guided audit workflow with step validation.
- Risk analysis + recommended next steps from audit history.
- **Success:** fewer invalid audits, stronger legal defensibility, less rework.

Team Cargo Flow — Pain Point & Proposed Solution

Pain Point

- Customs declaration backlog at borders.
- Slow clearance impacts trade throughput.
- Low-risk cargo treated like high-risk.

Proposed AI Solution

Team Members: Andiswa Skenjana, **Elsa van Rooyan (Lead)**, Elvis Mudzanani, Henk van der Heide

- ML trader profiling using compliance and transaction history.
- Anomaly detection on declarations + supporting documents.
- **Success:** faster clearance, targeted inspections, reduced border congestion.

Team AI G's (Gurus) — Pain Point & Proposed Solution

Pain Point

- Manual interpretation of cargo X-ray images.
- Repetitive work slows inspections.
- Inconsistent judgement risk in image review.

Proposed AI Solution

Team Members: Mpho Diko (Lead), Nkete Mashoene, Tebatso Mogale, Helario Nyandlale, Deshen Pillay

- Computer vision to classify X-ray patterns and flags.
- Match image signals vs declared goods + intelligence data.
- **Success:** faster inspections, consistent decisions, better detection precision.

Team Tax Avengers — Pain Point & Proposed Solution

Pain Point

- Rural access barriers: connectivity, literacy, language.
- No nearby branches + misinformation risk.
- Low trust reduces voluntary compliance.

Proposed AI Solution

Team Members: Azwifarwi Vele (Lead), Fikiswa Mdlalo, Fulufhelo Nemutandani, Khuthadzo Munyai

- Hybrid digital kiosks for underserved communities.
- Multilingual AI guidance for common SARS tasks.
- **Success:** improved inclusion, fewer errors, higher compliance participation.

SARS Co-Creation Synthesis — Key Pain Points

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- **Access Inequality:** Rural taxpayers face digital, linguistic, and trust barriers.

Global AI Case Studies — Revenue Administration

Estonia — AI-Enabled Digital Tax Ecosystem

AI Technologies

- AI-driven pre-filled returns
- Automated anomaly detection
- Real-time inter-agency data exchange
- Risk-based audit algorithms

Revenue Administration Impact

- 95%+ digital filing rate
- Filing completed in minutes
- High voluntary compliance
- Reduced audit overhead

United Kingdom (HMRC) — Connect AI System

AI Technologies

- Big data analytics platform
- Network analytics
- Machine learning fraud detection
- Cross-database intelligence matching

Revenue Administration Impact

- Billions recovered from tax gap
- Strong risk segmentation
- Reduced manual investigations
- Increased enforcement precision

China — AI-Powered Smart Taxation

AI Technologies

- AI-based invoice verification
- NLP taxpayer chatbots
- Predictive compliance modelling
- Real-time VAT fraud detection

Revenue Administration Impact

- Faster fraud detection
- Reduced VAT evasion
- Large-scale automation
- Improved taxpayer service

Brazil — AI in Electronic Invoicing & Compliance

AI Technologies

- National e-invoicing validation AI
- Real-time transaction monitoring
- Machine learning risk scoring
- Data mining for evasion networks

Revenue Administration Impact

- Greater VAT transparency
- Reduced invoice fraud
- Targeted enforcement
- Improved collection efficiency

Future Readiness, Research & Responsible AI

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- **Focus:** Foresight, governance, sustainability
  - Future tech + research capacity + responsible AI controls

Emerging Tech: Revenue Administration (Wave 1–2)

- **Wave 1: Predictive Revenue Forecasting**
  - AI macro-economic fusion models (multi-source signals)
  - Real-time sector risk modelling + scenario planning
  - Simulation-based budget stress tests
  - *Strategic question:* reactive collection vs predictive fiscal management

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- **Wave 2: Autonomous Compliance Systems**
  - Self-adjusting risk thresholds and prioritisation logic
  - Automated nudging *before* default (behavioural interventions)
  - AI-supervised human decisions (human-in-the-loop boundaries)
  - *Strategic question:* where does autonomy stop and judgement begin?

Emerging Tech: Revenue Administration (Wave 3–4)

- **Wave 3: Digital Identity Infrastructure**
  - Cross-platform identity validation (strong assurance)
  - Blockchain-backed refund integrity + traceable verification
  - Biometric integration options (where appropriate)
  - *Strategic question:* what is a fraud-resistant identity layer for SARS?

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- **Wave 4: Cross-Border Intelligence Integration**
  - Data-sharing architectures (secure, treaty-aligned)
  - International risk graph analytics (networks, beneficial ownership)
  - AI-enabled cooperation for complex compliance patterns
  - *Strategic question:* is SARS architecturally ready for global AI cooperation?

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The Future Readiness Question

Is AI a project at SARS,

or a permanent institutional capability?

Responsible AI: Risks in a Revenue Authority

- **Core Risk Domains**
  - Explainability & auditability (decisions must be defensible)
  - Bias & fairness (compliance targeting, segmentation, risk scoring)
  - Data governance & privacy (POPIA-aligned controls)
  - Cybersecurity exposure (model, data, identity, channel threats)
  - Regulatory compliance (process, records, accountability)

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- **Responsible AI Mindset**
  - We are not debating ethics — we are designing institutional controls
  - Every model needs ownership, monitoring, and a clear audit trail

Responsible AI: Institutional Controls (Minimum Set)

- **Five Controls (Practical & Implementable)**
  - **Model Registry:** inventory, purpose, data, versions, owners
  - **Bias Audit Cycle:** fairness metrics + periodic review
  - **Legal Defensibility Review:** explainability + evidence trails
  - **Cyber AI Shield:** monitoring + secure deployment patterns
  - **Ethics Oversight:** accountable approvals for high-risk use cases

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- **Non-Negotiable Outcome**
  - AI decisions must be transparent, auditable, and safe under scrutiny

From Workshop to Pilot: A Practical Pathway

- **Pilot Pathway Model**
  - Select 2–3 low-risk, high-value pilots (quick wins)
  - Define measurable success metrics (before build)
  - Assign an executive sponsor + accountable product owner
  - Run a 90-day evaluation cycle (iterate or stop)

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- **Capability Implications**
  - Data readiness, process redesign, skills, and compute requirements
  - Governance readiness: approvals, monitoring, audit trail standards

Ownership & Value: Making AI Sustainable

- **Governance Ownership Questions**
  - Who owns the model (business owner) and who owns the risk?
  - Who signs off deployment and monitors drift/bias?
  - Who is accountable when AI is wrong (process + escalation)?

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- **Value Measurement Framework (Minimum Set)**
  - Revenue impact (collection uplift, leakage reduction)
  - Risk reduction (fraud reduction, improved targeting)
  - Process efficiency (cycle time, throughput, rework)
  - Citizen trust signals (service quality + confidence indicators)

Programme Synthesis: Key Insights & Next Steps

- **Key Insights (What We Now Know)**
  - AI opportunities are real — but governance is the differentiator
  - Process design, decision quality, and trust move together
  - Capability must be institutionalised (skills + data + standards)

Programme Synthesis: Key Insights & Next Steps

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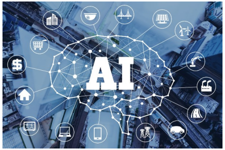
- **Closing Question**
  - What will SARS look like in 5 years if we do this properly?

Closure: SARS 2030 — AI as Institutional Capability



- **Three Commitments**
  - AI as a governed enterprise capability (standards + ownership)
  - AI as research-driven capability (learning, sandboxes, partnerships)
  - AI as ethically defensible public infrastructure (trust by design)

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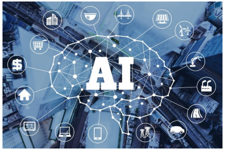

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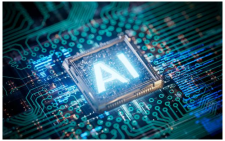
- **Final Output**
  - A shared roadmap: pilots, governance, capability build, measurement

Time Travel: Future of Tax Administration

Future of Tax Administration: Cash Societies



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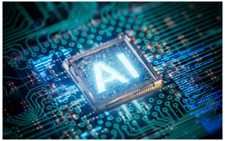

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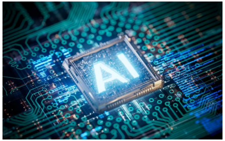

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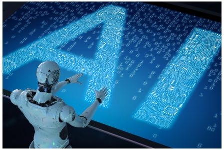
- **Strategic Tension**
  - Financial inclusion vs enforcement intensity

Future of Tax Administration: Blockchain & Programmable Economies



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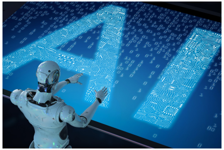

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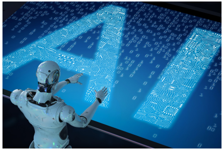

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- **Dual-Reality Future**
  - Managing untraceable cash flows
  - While regulating hyper-traceable blockchain economies

Reflection & Closing

Your Feedback on Day II — a word or phrase (Closed)

- Use one word, a phrase, or a short sentence to describe Day I.

Thank You — Please Let's Connect

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